DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination 2023

Course: First Year B. Tech. **Branch: Group A / Group B**

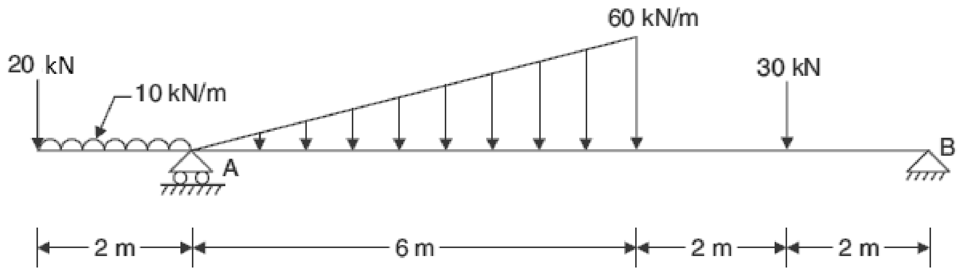
Subject Name: Engineering Mechanics **Subject Code: BTES103**

Max Marks: 60 **Date: 11/08/2023** **Duration: 3 Hrs.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

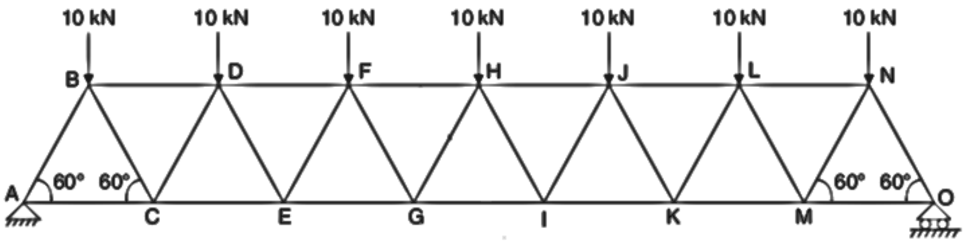
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	Classify the system of forces with near sketches & explain them?	**Remember**	**06**
B)	A beam AB of length 5 m supported at A and B carries two point loads W_1 and W_2 of 3 kN and 5 kN which are 1 m apart. If the reaction at B is 2 kN more than that at A, find the distance between the support A and the load 3 kN.	**CO2**	**06**
C)	Find the magnitude and direction of the resultant of the concurrent forces of 8 N, 12 N, 15 N and 20 N making angles of 30°, 70°, 120°.25 and 155° respectively with a fixed line.	**CO2**	**06**
Q. 2	**Solve Any Two of the following.**		
A)	State and prove in detail Varignon’s theorem with example?	**Remember**	**06**
B)	Overhanging beam shown in figure is on roller at end A and is hinged at the end B. Determine the reactions developed for the loadings shown in figure.	**CO2**	**06**



C)	An I-section has the following dimensions in mm units: Top flange = 150 x 50 Web = 50 x 300. Bottom flange = 300 x 100 Determine the position of Centroid of the section.	**Application**	**06**
Q. 3	**Solve Any Two of the following.**		
A)	State and prove parallel axis theorem of moment of inertia.	**Remember**	**06**

B) An effort of 200 N is required just to move a certain body up an inclined plane of angle 15° the force acting parallel to the plane. If the angle of inclination of the plane is made 20° the effort required, again applied parallel to the plane, is found to be 230 N. Find the weight of the body and the coefficient of friction. **CO2** **06**

C) Determine the forces in the members FH, HG and GI in the truss shown in figure. Each load is 10 kN and all triangles are equilaterals with sides equal to 4 m. **CO2** **06**



Q. 4 Solve Any Two of the following.

A) A small steel ball is shot vertically upwards from the top of a building 25 m above the ground with an initial velocity of 18 m/sec. **CO 5** **06**

a) In what time, it will reach the maximum height?

b) How high above the building will the ball rise?

B) A wheel increases its speed from 45 *r.p.m* to 90 *r.p.m* in 30 sec. Find i) angular acceleration of the wheel and ii) No. of revolutions made by the wheel in these 30 sec. **CO 4** **06**

C) The horizontal component of the velocity of a projectile is twice its initial vertical component. Find the range on the horizontal plane, if the projectile passes through a point 18 m horizontally and 3 m vertically above the point of projection. **CO 4** **06**

Q. 5 Solve Any Two of the following.

A) State and prove work energy principle. **Remember** **06**

B) A motorist travelling at a speed of 70 kmph suddenly applies brakes and halts after skidding 50 m. Determine: **CO 5** **06**

(1) the time required to stop the car;

(2) the coefficient of friction between the tyres and the road.

C) Direct central impact occurs between a 300 N body moving to the right with a velocity of 6 m/s and 150 N body moving to the left with a velocity of 10 m/s. Find the velocity of each body after impact if the coefficient of restitution is 0.8. **CO 5** **06**

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